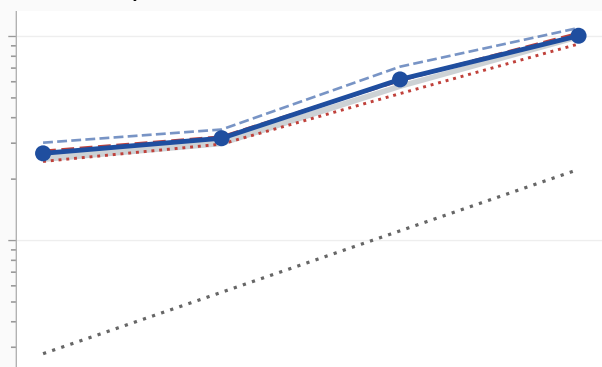


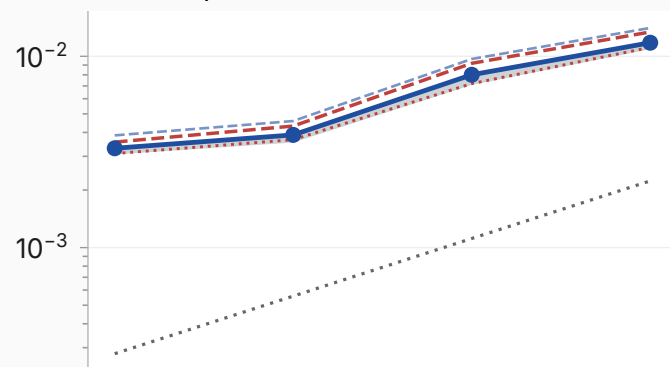
Does the source context vector live in the top-k key subspace?
 projection energy onto the top-k right-singular subspace of the stacked q/k/v update, band mean L14-L24, module-input space

context-vector energy captured

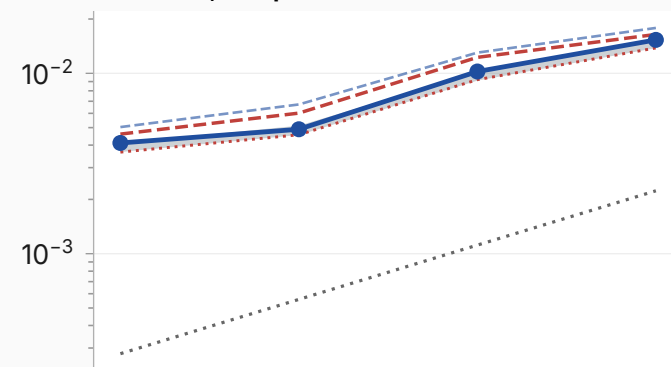
dose dial, shallow window



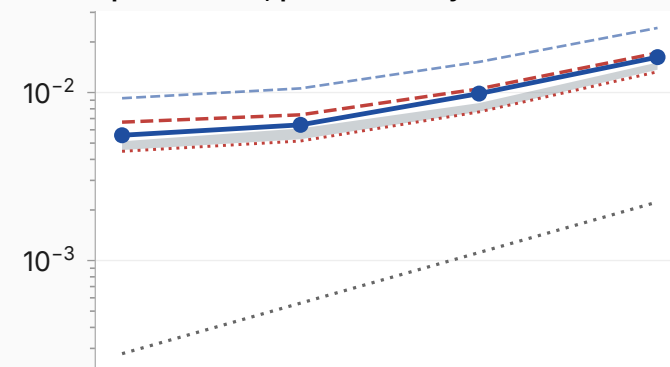
dose dial, mid window



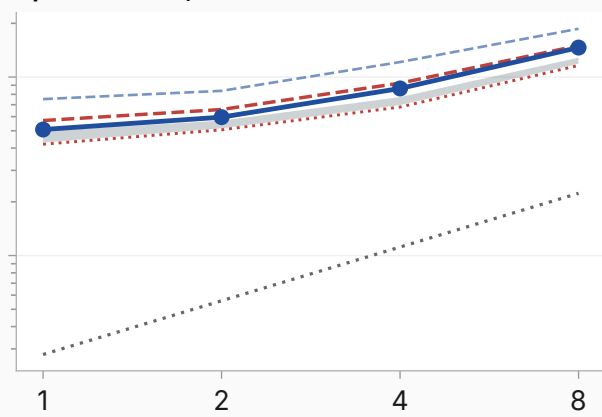
dose dial, deep window



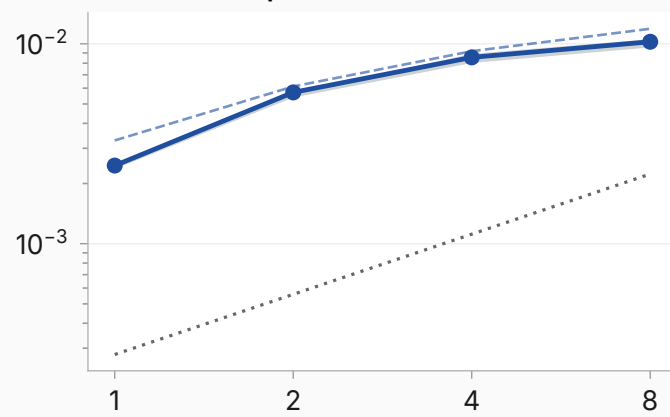
epoch ladder, positives-only



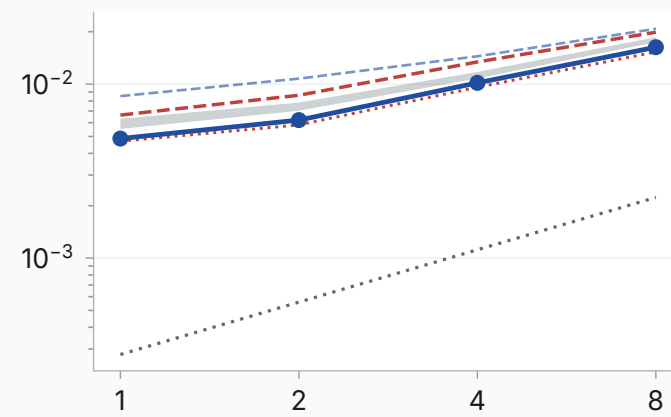
epoch ladder, contrastive



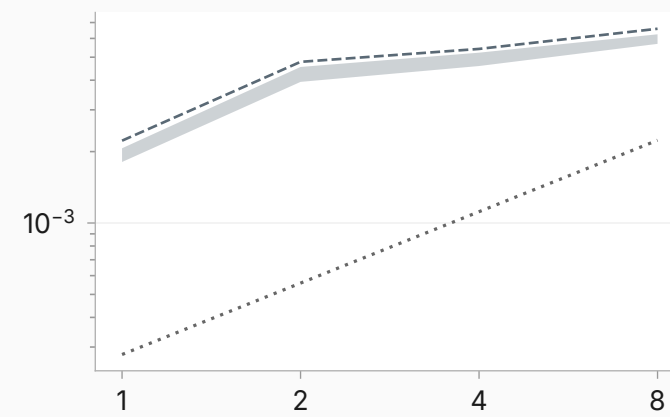
saturated endpoint



cross-behavior



EM no-source control (scale ref)



key-subspace size k

key-subspace size k

key-subspace size k

key-subspace size k

wrong-context null (p50-p95) matched source (median)
 shuffled-pairing null p95 matched source (max cell)
 shuffled-pairing null p50 random-vector floor $k/3584$

whole-bank read (p50-p95)
 whole-bank max